

Architectural Particular Specification

STEEL AND METAL WORKS

SECTION 4.1

Rev	Date	Description
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SECTION 4.1

STEEL AND METAL WORKS

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The Work of this Section includes, but is not limited to, the furnishing and installation of the following :
1. Load bearing formed steel stud external and internal framing.
 2. Non-load-bearing cold formed internal metal stud wall and ceiling framing.
 3. Steel pan stairs (permanent formwork).
 4. Fasteners, framing hardware and other ancillary materials.
 5. Structural Engineering design, calculations and shop drawings for any load-bearing metal framing.
 6. Touch-up painting of welded connections in metal framing.
 7. Shop fabricated ferrous and non-ferrous metal items, galvanizing, priming and painting.
 8. Miscellaneous metal fixings, items, etc., embedded into cast-in-place concrete.
 9. Connection of miscellaneous framing and supports to structural steel.
 10. Galvanic corrosion protection and isolation of dissimilar metals and incompatible substrates.
 11. Provide ornamental metals in accordance with the Contract Documents
 12. The supply installation and shop fabrication of all exposed exterior aluminium sheets and metalwork as noted herein and on the drawings. The Contractor shall provide and install all necessary fixings, bracing and supports as required for a complete installation of the architectural metalwork
 13. Fabrication of metal sheets involving, but not limited to, forming, bending, brake forming, acid-etching, embossing, blanking, perforating, piercing and welding
 14. Counter flashing at all plumbing, electrical, lighting protection, mechanical systems, equipments, vent stacks and other roof and wall penetrations.
 15. Temporary fire protection measure of flashing and sheet metal Work during installation of all torch-on applied modified bitumen roofing Work.
 16. Organic coating of architectural metals, galvanizing, priming and painting.
 17. The design of secondary steelwork metalwork support to the various items within the scope of work and the design of items specified within the Building (Construction) Regulations Section 17.
 18. Coordinate and interface with all Roofing Work to ensure weather tightness and warrantable work.
 19. Coordination with sub-contractors, particularly with respect to the provision of power to light fixtures mounted on metalwork.

1.02 REFERENCE

(NOT USED)

1.03 QUALITY ASSURANCE

(NOT USED)

1.04 PERFORMANCE REQUIREMENTS

- A. Materials shall be best quality obtainable.
- B. Materials shall be clean, straight and free from twist, buckle or other defects before fabrication and assembly. If straightening and flattening is required, it shall be done in a manner that will not injure the material.
- C. Irrespective of the materials and articles offered in the tender, all materials and articles for use of the work shall be in full compliance with the Specification unless otherwise agreed or approved by the Architect.
- D. Any fabrication conducted outside of Hong Kong Special Administrative Region shall be required to comply with the standards within this specification including but not limited to all British Standards and Code of Practice, circulars, guidelines and specifications issued by the relevant government departments.

1.05 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

1.07 SUBMITTALS

- A. Shop Drawings and Samples
 - 1. Provide shop drawings where required by the Particular Specification or drawings.
 - 2. Shop drawing shall show full details of the components to be fabricated, finishes and the method of fixing on site.
 - 3. Drawings shall include weld information and weld symbols conforming to B.S. 499. Fixing that will be visible in the completed work shall be clearly indicated on drawings.
 - 4. Where required, drawings shall be supported by structural calculations. Calculation shall be clearly referenced to and coordinated with shop drawings.
 - 5. Samples of metalwork and sections, including all finishes shall be supplied as requested by the Architect for approval prior to fabrication. All metalwork shall be fabricated and finished to the same standard as approved samples.
 - 6. The Contractor shall prepare and submit 7 sets of detail shop drawings (including calculations) for Architect's confirmation of no adverse comment. Note that Architect's drawings illustrate design intents and are diagrammatic. The design calculations shall include a synopsis stating the relevant codes and standards, the design loadings and the design consideration, and relevant calculations that substantiate all curtain walling types and their connection and anchorage satisfy the requirements of strength and stiffness under both the working and the test load conditions

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY STORAGE AND HANDLING

- A. Deliver metal work fully sealed and identified. Protect from damage from any source. Provide removable protection as required.
- B. Store indoors, above the floor, protected from construction activities and other sources of damage.

1.10 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

(NOT USED)

1.12 WARRANTY

This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCT

2.01 APPROVED MANUFACTURER

(NOT USED)

2.02 MATERIALS / SYSTEMS

A. Cast Iron

1. Cast iron shall be grey cast iron of the finest quality complying with B.S. 1452 : 1990 : Grade 150.
2. Castings shall be:-
3. Sharp and sound,
4. Of the exact form required, shaped to fit the parts truly and to hold full dimensions, and
5. Entirely free from air holes, scoriae, core nails flaws, cracks, voids and defects of every kind.
6. Holes shall be drilled as required for testing the thickness of the metal and filled up afterwards, all at the Contractor's expense.
7. Sea sand shall not be used in casting.

B. Mild Steel Sections

1. Mild steel hot-rolled sections shall be made from weldable structural steels to B.S. 7613 : 1994 and B.S. 7668 : 1994.
2. Hot rolled sections (Except angles) shall comply with B.S. 4 : Part 1 : 1993.
3. Hot-rolled hollow sections shall comply with B.S. 4848 : Part 2 : 1991.
4. Hot-rolled angles shall comply with B.S. 4848 : Part 4 : 1972 (1986).
5. Flat rolled carbon steel plates, sheets and strips shall comply with B.S. 1449 : Part 1 : 1991.
7. Cold rolled steel sections shall comply with B.S. 2994 : 1976 : (1987).
8. Do not use cold rolled steel sections without the prior approval of the Architect.

C. Mild Steel Plates, Rods and Bars

1. Mild steel plates, rods and bars shall be:-
2. Of low carbon type mild steel,
3. Perfectly straight,
4. Free from winding except where otherwise shown, and
5. In one length.
6. Mild steel plates, rods and bars shall be bent to exact forms required and the ends filed to form true joints.

D. Galvanized Steel Tubes, Sheets and Coils

1. Galvanized steel tubes and tubulars shall comply with B.S. 1387 : 1985 (1990), Medium Grade.
2. Hot-dip zinc coated and alloy coated steel sheets and coils shall comply with B.S. EN 10143 : 1993.

E. Slotted Steel Angle

1. Slotted steel angle shall be self-finished angle to B.S. 4345 : 1968 (1986).
2. Fittings shall be brass or cadmium-plated steel.

F. Corrugated Steel Sheet

1. Corrugated steel sheet shall be hot-dip zinc coated steel complying with B.S. 3083 :1988.
2. Corrugated steel sheet shall be:-
 - a. Of tough mild steel, well annealed, even in temper and thickness, free from holes, cracks, blisters and other defects.
 - b. Perfectly square,
 - c. Having the corrugations parallel with the sides, and
 - d. Regular in curve, pitch and depth.
3. Unless otherwise specified, corrugations shall be of standard pattern, 76mm wide and 19mm deep.

G. Stainless Steel

1. Stainless steel plates, sheets and strips shall be 18/9 chromium nickel steel complying with B.S. 1449 : Part 2 : 1983 of:-
 - a. Finish No. 2B for mill finish,
 - b. Finish No. 2A for reflective finish,
 - c. Finish No. 4 for polished finish,
 - d. Finish No. 8 for bright polished finish with a high degree of image clarity.
2. The latter one is to be corresponding to Japanese quality grade 316 S16 (to be used externally or in semi-external spaces) and grade 304 S16 (to be used internally).
3. Stainless steel welded tubes for structural and general engineering purposes shall comply with BS 6323: Part 1 & 8: 1990 composition and finish as above.

H. Copper and Copper Alloy

1. Bronze shall be an alloy of:
 - a. 56.5% copper
 - b. 41.45% zinc
 - c. 2.25% lead
2. and shall have a natural colour polished finish unless specified otherwise.
3. Brass shall be an alloy of the following type:-
 - a. Alpha brass 0 - 37% zinc content
 - b. Alpha plus beta brass 37 - 46% zinc content
 - c. Beta brass 46 - 50% zinc content
4. Rolled copper and copper alloy sheets, strips and foils for roofing, building and decorative purposes shall comply with the general requirements of B.S. 2870 : 1980.
5. Copper and copper alloy round tubes for general purposes shall comply with B.S. 2871 : Part 2 : 1972 of the required grade.
6. Copper and copper alloy rods and sections for general engineering purposes shall comply with B.S. 2874 : 1986 of the required grade.

7. Wrought copper and copper alloy plates shall comply with B.S. 2875 :1969.
8. Brass rods and sections shall be designation C2106 condition M.
9. Consolidated bass shall comply with B.S. 6722 : 1986 (1992) Part 1 & 2.

I. Aluminum Alloy

1. Structural aluminium alloy section shall comply with B.S. 1161 : 1977 (1991).
2. Aluminium alloys bars, extruded round tubes and sections shall comply with B.S. 1474 : 1987 alloy designation 6063.
3. Aluminium alloy plates, sheets and strips shall comply with B.S. EN 485 of the required alloy designation.
4. Aluminium alloy drawn tubes shall comply with B.S. 1471 : 1972 alloy designation 6063.
5. Aluminium units shall be:-
 - a. Framed with corners of adequate strength and design to prevent distortion,
 - b. Reinforced with steel as necessary and with hardwood cores where required and,
 - c. Factory welded, riveted and bolted.
 - d. All bolts and screws for aluminium units shall be :-
 - e. Of stainless steel or alloy endorsed by the Architect,
 - f. Countersunk flush where practicable,
 - g. Semi-countersunk round headed where not practicable to countersunk fully.
6. Aluminium sections for structural purposes to be to B.S. 1161 : 1979 (1991).

J. Steel Lathing and Expanded Metal

1. Steel lathing for plastering and rendering shall comply with B.S. 1369 : Part 1 : 1987 (1994).
2. Expanded metal carbon steel sheets for general purposes shall be "Expamet" to B.S. 405 : 1987.
3. Expanded metal grating panels for use in flooring, walkway and stair treads shall comply with B.S. 4592 : Part 2 : 1987 (1994).
4. Expanded metal lathing shall be "Expamet" produced by the Expanded Metal Co. Ltd. or other equal and approved and of the following types:
 - a. For use with suspended ceilings of cement render, tile or plaster, Expamet Rib Lath 271 for bearer spacing of 600mm.
 - b. For use in 300mm strips for strengthening cement render where shown on the drawings and at expansion joints, Expamet BB263. The strips are to be cut so that the maximum stiffness is obtained across the strip. The strip is to be fixed to the concrete or brick at one edge and at the centre, at not more than 300mm centres.
 - c. For bonding brickwork to Expamet BB 249, 300mm long at 450mm centres vertically.
 - d. For general screed reinforcing use Expamet BB 263.
 - e. For reinforcement of brick or concrete block walls, Expamet strip reinforcing in width 12mm less than the width of the wall, laid in courses at 450mm centres to centre vertically.
 - f. For open mesh walkways, Expamet WK 4894.

5. All metal lathing shall be fixed as taut as possible with largest dimension of the mesh running between supports and with stiffening ribs uppermost in the same direction. Cut ends shall be turned away from the plastered surface.
6. End laps shall be not less than 25mm when the lap occurs at the bearer and otherwise not less than 50mm. Side laps are to be not less than 25mm. All laps are to be securely wired together with 18 gauge soft galvanized binding wire at intervals of not more than 75mm.

K. Wire Netting

1. Hexagonal steel wire netting shall be zinc coated, manufactured by mild steel wire to B.S. 1052 : 1980 (1986) and shall comply with B.S. 1485 : 1983 (1989).
2. Zinc coated woven wire mesh for hinged jointed fencing shall comply with Section Five of B.S. 4102 : 1990.

L. Steel Wire Mesh

1. Welded wire mesh shall be steel wire of the specified diameters welded to form a square or oblong mesh of the required mesh aperture size as required and shall be hot dip galvanized after manufacture if required.
2. Welded steel fabric shall be "BRC Fabric" mesh. Fabric for concrete reinforcement shall comply with B.S. 4483 : 1985.
3. Galvanized welded steel wire mesh shall comply with B.S. 4102 : 1990 Section Six.
4. High tensile steel wire mesh with square apertures shall comply with B.S. 7372 : Part 3 : 1993.
5. Wire mesh for reinforcing screeds shall be galvanized, 1.5mm diameter wire mesh not larger than 20mm.

M. Welding Electrodes

1. Electrodes for manual metal-arc welding shall comply with B.S. 499 : 1995.
2. Filler rods for gas welding shall comply with B.S. 1453 : 1972 (1987).
3. Filler metals for brazing shall comply with B.S. 1845 : 1984.
4. Filler rods for gas-shielded arch welding shall comply with B.S. 2901 : 1990.

N. Fixing Generally

1. Unless otherwise specified, all fixings shall be of the same materials and finish as the materials to be fixed.

O. Rivets

1. Rivets for general engineering purposes shall be mild steel complying with B.S. 4620 : 1970 (1988).

P. Bolts, Screws, Nuts and Plugs

1. Black hexagon bolts, screws and nuts shall comply with B.S. 4190 : 1967.
2. High strength friction grip bolts and associated nuts and washers for structural engineering shall comply with B.S. 4395 : Part 1 : 1969 and Part 2 : 1969.
3. The use of high strength friction grip bolts in structural steelwork shall comply with B.S. 4604 : Part 1 : 1970 and Part 2 : 1970.
4. Machine screws and nuts shall comply with B.S. 4183 : 1967.
5. Self-tapping screws shall comply with B.S. 4174 : 1972.
6. Rivets shall be as follows:

- a. solid rivets to B.S. 4620 : 1970 (1988).
 - b. bifurcated rivets to B.S. 4894 : 1993.
 - c. semi-tubular rivets to B.S. 4895 : 1981.
7. Plugs shall be proprietary fibre or plastic plugs suitable for the purpose.

Q. Fixing Bolts

- 1. Stud anchors and fixing bolts shall be approved proprietary expanding bolts equal to:-
 - a. Phillips Redhead” or,
 - b. “Rawlbolts“.
- 2. Foundation bolts and nuts shall conform to B.S. 4185 : Part 9 : 1976 (1990).
- 3. All fixing bolts shall be of sizes as specified and used in accordance with the manufacturer’s recommendations.

R. Metal Insert Channels for Concrete

- 1. Metal insert channel for concrete shall be an approved proprietary galvanized sheet channel filled with a removable expanded polystyrene temporary filler.
- 2. Insert channels shall be supplied complete with proprietary and matching sliding fixings as required.

S. Chain Link Fence

- 1. Steel wire for fencing including chain link and barbed wire shall comply with B.S. 4102 : 1990 Section One, zinc coated or plastic coated (Grade A or B) as specified.
- 2. Chain link fences shall comply with B.S. 1722 : Part 1, B.S. 1722 : Part 10 and B.S. 1722 : Part 13.

T. Sink Brackets

- 1. Provide and fix at least two (2) nos. of brackets for the support of each stainless steel sink units.
- 2. Sink brackets shall be:-
 - a. Constructed with 40 x 40 x 6mm thick mild steel angle sections of suitable length,
 - b. Built into wall and,
 - c. Drilled and countersunk to receive fixing screws.

U. Shelf Brackets

- 1. Shelf brackets, unless otherwise specified, shall be formed by mild steel angle sections as follows:-
 - a. Section of flange size equal to the thickness of the shelf and,
 - b. Length of shelf brackets equal to the depth of the shelf,
 - c. Brackets shall be fixed at maximum 600mm centres. Shelves with a depth greater than 300mm shall be supported as detailed on drawings.

V. Dowels

- 1. Dowels to door frames, stud partition frames, etc. shall be:-
 - a. Of 12mm diameter mild steel rod 75mm long.

- b. Mortised for a depth of 40mm into bottom of door frames jambs, etc. with a tight push fit, and
- c. Other end bedded in concrete floor and grouted with neat cement mortar.

W. Holdfasts

1. Provide and fix, six (6) nos. of holdfasts for each door frame (Three (3) nos. for each jamb).
2. Holdfasts for door frames to masonry wall shall be as follows:-
 - a. Of galvanized mild steel flat 40 x 3 x 200mm girth with 50mm turned up at one end.
 - b. Twice drilled and countersunk for screws of not less than 6mm diameter to backs of door frames, and
 - c. The other end split and fishtailed for building in.
3. Holdfasts for door frames to reinforced concrete wall or column shall be provided with six (6) nos. 12mm diameter "HILTI-HT" expanding anchor bolts 132mm long overall, with nut and washer including necessary tightening and pelleting.

X. Adhesive for Metal

Adhesive for bonding metal to wood or metal to metal shall be an approved proprietary brand.

Y. Bitumen Coating Solution

Bitumen coating solution for cold application shall comply with B.S. 3416 : 1991 : Type 1.

Z. Sundries

1. Fire protection building board lining
 - a. Building boards for fire protection lining shall be asbestosfree non-combustible building boards.
 - b. Fire protection lining shall be of fibre-cement composite panels reinforced with 0.5mm thick hot dip galvanized steel sheets mechanically bonded on both sides, and shall be of the following nominal thicknesses, sizes and approximate weights.

THICKNESS	SIZE		APPROX. WEIGHT	
	WIDTH	LENGTH	(kg / m ²)	(lb / ft ²)
6.0mm	750mm or 900mm or	1815mm or 2425mm	16.8	3.44
9.5mm	750mm or 900mm or	1815mm or 2425mm	21.0	4.30

- c. Building boards used for fibre protection lining shall be fibre silicate building boards and shall be of the following nominal thicknesses, standard sizes and approximate weights:-

THICKNESS	SIZE	APPROX. WEIGHT	
		Dry	With 6% moisture
6mm	1.25 x 2.5m	5.3 kg/m ²	5.6 kg/m ²
8mm	1.25 x 2.5m	7.0 kg/m ²	7.4 kg/m ²
10mm	1.25 x 2.5m or 3.00m	8.7 kg/m ²	9.2 kg/m ²
12mm	1.25 x 2.5m	10.5 kg/m ²	11.1 kg/m ²
15mm	1.25 x 2.5m or 3.00m	13.1 kg/m ²	13.9 kg/m ²

- d. Unless otherwise specified or shown on drawings, "DURASTEEL" composite panels or "PROMATECT-H" building boards used in forming fire protection enclosure, casing, lining to cable and ventilation ducts, partitions, suspended ceiling, etc., shall be so constructed and of the thickness as recommended by the manufacturer and in accordance with the requirements of fire rating as specified and fixed to background with and including provision of 50 x 50 x 5mm mild steel angle section framed supports using M8 self-tapping screws at 300mm (maximum) centres.
- e. The Contractor shall submit test certificates from the manufacturer and letters of approval from the Government recognizing that the said construction and materials comply with the requirements of B.S. 476 : Part 20 : 1987.

2. Access and Panels and Access Doors

Unless otherwise specified, access panels and doors shall be fabricated from mild steel units using 38mm mild steel angle frame and mild steel panels. Minimum thickness of mild steel panels to be 3mm. Access panels and doors shall be supplied complete with locking device and all other fittings shown on drawing.

3. F.S. / Sprinkler / Inlet Cabinets

All inlet cabinets shall be made from steel, aluminium or stainless steel as detailed on drawings and in compliance with the Code of Practice for Minimum Fire Service Installations and Equipment.

4. Galvanized Steel Tubular Handrails

- a. Unless otherwise specified, galvanized steel tubular handrails for internal fixing shall be 38mm internal diameter galvanized steel tubing cut to lengths and capped at both lengths.
- b. Carefully notch, fit and weld tubing to produce accurate joints which shall be ground smooth and flush and finished with two coats of zinc rich priming paint to B.S. 4652 : 1995 : Type 2.
- c. Brackets shall be fabricated from 6mm thick, 32mm wide flat mild steel bar welded to the tubular rail, finished as described above and fixed to concrete or masonry walls at 900mm centres approximately.

5. Mild Steel Cores for Vinyl Covered Handrails

- a. Unless otherwise specified, provide and fix 6mm thick, 38mm wide flat mild steel cores for vinyl cover and to give a smooth finished surface.

6. Galvanized Mild Steel Ladders
 - a. Provide mild steel step ladders to all water storage tanks and lift pits. Unless otherwise specified, ladders shall be constructed from 13mm thick, 38mm wide galvanized mild steel strings with 19mm diameter mild steel rungs at 300mm centres. The strings shall be continued above the top of the tank or roof surface, or to be bent or fish-tailed at ends to allow ends to be cast-in situ fixed to concrete or built-in to masonry.
 - b. Cat ladders leading to upper roof level shall be fitted with a cage door 1200mm high, fabricated with 38mm mild steel angle frame, 1.6mm thick mild steel panel and complete with locking device.
7. Step Irons
 - a. Step irons shall be galvanized mild steel, stainless steel, aluminium or plastics encapsulated steps as specified and to B.S. 1247 : Parts 1 : 1990, Part 2 : 1990 & Part 3 : 1991.
 - b. Step irons shall be cast in or built in and fixed at 300mm centres, alternate steps staggered.
8. Slotted Steel Angle
 - a. Cut slotted steel angle square. Securely bolt together using nuts, bolts, washers and angle braces as required.
9. Exit Sign Boxes

Provide and install exit sign boxes in accordance with the detail drawings and in compliance with the related British Standards.
10. Chain Link Fence

Provide and install chain link fencing in accordance with the drawings.
11. Shower Curtain / Towel Rods

Unless otherwise specified, shower curtain rails and towel rods shall be 25mm Dia. 3mm THK. stainless steel heavy duty hollow tube rose fixed, with matching purpose made end flanges stainless steel to be 18/9 chromium nickel steel.
12. Fan Hooks

Fan hooks shall be formed to "S" shape with 12mm diameter brass rods and built in where required. Built in end to be hooked around reinforcing rod of the floor from which they are suspended.
13. Position for Equipotential Bonding

Provide all external metalwork with screw holes or other connection points to the approval of the Architect for the purposes of connection, by the Electrical Installation Sub-Contractor, of equipotential bonding conductors as required by the I.E.E. Wiring Regulations.
14. External Aluminium Cladding Panels

Refer to Section 9.1 of Particular Specification for Aluminum Windows, Window Walls, Glass Canopy, Skylight, Aluminum Louvres & Features.
15. Internal Aluminium Cladding Panels

The aluminium panel system shall include without being limited to the following:

 - a. Aluminium cladding of no less than 3mm in thickness coated with PVF2 coat and stiffened with extruded aluminium stiffeners bolted to the back of cladding sheet via welded aluminium studs.
 - b. Extruded aluminium holding tracks of no less than 3mm in thickness with exposed surface coated with PVF2 coat of colour as specified by the architect.

- c. High impact plastic isolator shall be provided at all cladding panels and/or fixing tracks to prevent movement noises of cladding.
 - d. Extruded aluminium or stainless steel bracketary for connecting the holding track to the building structure.
 - e. Stainless steel anchorage for locations beyond vapour barrier.
 - f. Aluminium cladding shall be manufactured and fabricated including cut, notched, bent, folded, welded and coated at the proprietary manufacturer's plant and the proprietary cladding manufacturer shall be proposed to and approved by the Architect.
16. The name of the manufacturer of aluminium panel shall be submitted and approved by the Architect prior to the award of the contract. The source must have a satisfactory proven track record of aluminium panels supplied for the type, size and thickness concerned, appropriate QA/QC measures, and compliance with best practice in aluminium panel manufacturing. The fabrication and coating of the panels shall be carried out by the same manufacturer and no fabrication or modification work carried out at the Contractor's in-house fabrication facility shall be allowed.
 17. Panel Manufacturer shall be responsible for providing all product guarantees for project aluminium panels.
 18. All V-cut corners and joints of the panels shall be TIG-welded fully along the joints. All welding shall be ground down and sanded down with Grade 180 abrasive paper to a smooth finish prior to coating process.
 19. All fabricated panels shall be pre-treated prior to coating. The pre-treatment shall be by immersion method in a minimum of 12 stage process which includes degreasing, etching, desmutting and chromate stages. The final rinse shall be with de-ionized water.
 20. All coatings shall be applied after fabrication of panels. Pre-coated materials shall not be used. The curing of the coatings shall be by conveyORIZED oven to ensure consistency of curing temperature.
 21. Aluminium panel cladding shall be designed to a maximum deflection.
 22. All panel cladding shall be formed from a minimum thickness of 3mm aluminium alloy 3003-H14 or 5005-H34 and finished after fabrication. Submit mill certificate for approval. Exact thickness shall be as required for both structural and visual considerations. All exposed exterior components of the panel system shall be designed, fabricated and installed in such a manner that they produce a visual flatness when viewed from any angle.
 23. The maximum deviation of any point from its theoretical mathematical position shall be as follows:
 - a. Panels of exposed surface areas up to 3mm²:
1mm
 - b. Panels of exposed surface areas of 3mm² to 9mm²:
2mm
 - c. Panels of exposed surface areas exceeding 9mm²:
3.2mm
 24. The aluminium panels shall be installed and erected to comply with the following:
 - a. Deviation from plumb, level or dimensioned angle shall not exceed 1mm per 1000mm of length of any member and shall not be accumulative.
 - b. Maximum offset from true alignment between two consecutive members placed end to end shall not exceed 1.5mm.
 - c. Maximum offset between glass framing members at corners of glazing pocket shall not exceed 0.8mm.

25. Panels larger than 2500mm in any one dimension shall be backed with 9mm thick calcium silicate matrix reinforced fibre board. The anti-drumming backing shall be glued with proprietary contact adhesive approved by the Architect to the back of the flat and column claddings. The adhesive shall resist heat and continuous load stress and shall comply with MIL-A-21366A and MMM-A-121.
26. To prevent filiform corrosion, the contractor shall be required to take the utmost care in protecting the chromate conversion coatings from damage during coating operations, fabrication and installation. Such protections shall include but not be limited to:
 - a. Protect panel ends from damage, such as contact with concrete surfaces after coating.
 - b. Application of film coating.
 - c. Filiform corrosion resistance testing.
 - d. Selection of packing materials to prevent abrasion in shipping.
 - e. Care in storage, handling and erection.
 - f. Prompt inspection of installed panels.
 - g. Prompt post-erection repair of all breaks in the coating surface including cleaning, pre-treatment of the metal substrate in compliance with manufacturer's recommendations, and re-coating as recommended by coating system manufacturer.

2.03 ANCILLARY MATERIALS

(NOT USED)

2.04 FABRICATION

(NOT USED)

2.05 FINISHES/FINISHING TREATMENTS

(NOT USED)

2.06 STORAGE

(NOT USED)

2.07 WORKMANSHIP

(NOT USED)

PART 3 EXECUTION

3.01 EXAMINATION / INSPECTION

- A. Verify that field conditions are acceptable and are ready to receive work.
- B. Discrepancies: Do not proceed with installation in areas of discrepancies until such discrepancy has been fully resolved.
- C. Coordinate the Work of this Section with applicable trades.
- D. Should welding test results indicate that welds are below the standard of acceptance, carry out approved remedial measures and further acceptance tests. Retesting and remedial measures shall be at the expense of the Metal Fabricator.
- E. For ornamental metal works, examine the substrates, adjoining construction and conditions under which the Work is to be installed. Do not proceed with the Work until unsatisfactory conditions have been corrected.

3.02 PREPARATION

- A. Clean and strip primed steel items to bare metal where site welding is required.
- B. Supply setting templates to the appropriate entities for steel items required to be cast into concrete or embedded in masonry.
- C. Obtain Architect's approval prior to site cutting or making adjustments not scheduled.
- D. Clean and strip site primed steel items to bare metal where site welding is scheduled.
- E. Use non-shrink grout specified in the Structural Drawings and Specification for setting metal fabrications.

3.03 INSTALLATION / APPLICATION

- A. Fabricating Generally
 - 1. Fabricate metalwork with due care and accuracy and in compliance with the design and performance requirements, using a type and grade of metal, or metals, fit for the purpose.
 - 2. Before starting fabrication of repetitive components, obtain approval of prototypes from the Architect.
 - 3. The design of metal articles to be coated shall be in accordance with the recommendations of B.S. 4479 : 1990.
 - 4. During fabrication, protect all surfaces which will be visible in the finished work.
 - 5. Junctions of identical sections shall be mitred unless otherwise specified.
 - 6. Prepare surfaces of metal to be adhesive bonded, degreasing and mechanically or chemically abrading surfaces as recommended by the adhesive manufacturer. Adhesive shall be used to manufacturer's recommendations.
 - 7. Moving parts shall be assembled to move freely and without binding.
 - 8. Riveted joints are to be drawn tightly together with rivets fully clenched in, and over drilled holes.
 - 9. All burrs and sharp arrises which will be visible after fixing or a hazard to the user shall be removed.

B. Welding

1. All weldings of mild steel shall be arc welding carried out in accordance with B.S. 5135 : 1984.
2. Remove grease, dirt, moisture, oxide and scale from edges to be welded.
3. Ensure accuracy, using clamps or jigs where practicable.
4. Use tack welds for temporary attachment where jigs not practicable.
5. Joints to be fully fused throughout, with no holes, pores or cracks.
6. Prevent weld spatter falling on high tensile steel and self-finished surfaces visible in the completed work.
7. Ensure complete removal of flux residue and slag.
8. Welds visible in completed work to be as follows :-
 - a. Butt welds to be ground smooth and flush,
 - b. Fillet welds to be ground smooth.
9. Welding of steel shall be:-
 - a. gas welding to B.S. 2640 : 1982.
 - b. metal-arc welding to B.S. 5135 : 1984.
 - c. projection welding to B.S. 2630 : 1982 (1991).
 - d. seam welding to B.S. 6265 : 1982 (1991), or
 - e. other method subject to approval.
10. Welding of stainless steel shall be:-
 - a. inert-gas tungsten-arc welding carried out in accordance with B.S.. 7495 : 1991, or
 - b. other method subject to approval.
11. Welding of aluminium alloy shall be:-
 - a. inert-gas tungsten-arc welding carried out in accordance with B.S. 3019 : Part 1 : 1984.
 - b. inert-gas metal-arc welding to B.S. 3571 : Part 1 : 1985.
 - c. other method subject to approval.
12. Welding of copper and copper alloys shall be:-
 - a. bronze welding to B.S. 1724 : 1990 (1995), or
 - b. other method subject to approval.
13. Brazing shall comply with B.S. 1723.

C. Rivet and Bolt Holes

1. Rivet and bolt holes shall be drilled or reamed out to the exact diameter required and the burrs cleaned off.
2. Holes shall be set out to correspond exactly with the opposite holes in other parts to which the rivets and bolts have to be connected.
3. Holes shall be countersunk where required, the countersinking shall be concentric.

D. Riveting

1. All rivets shall be firmly driven so as to completely fill the holes and the heads shall be neatly finished off.

2. Bolting
 - a. All bolts shall have the length projecting through nuts by at least one thread, and shall have hexagon heads and nuts.
 - b. All bolts shall be fitted with washers which are to be tapered if necessary to give a level and even bearing on the nuts.
3. Joints
 - a. Riveted joints to be drawn tightly together, with rivets completely filling holes.
 - b. Mechanical joints to be tight, with no visible gaps.
4. Smith's Work and Fixing
 - a. Straightening and flattening shall be done in such a manner that will not injure the material.
 - b. All Smith's work shall be:-
 - 1) Forged clean from the anvil with flatters, swages or rounding tools and,
 - 2) Neatly chamfered at the edges or diminished as required.
 - c. All welds, tenons and sets shall be soundly made and all superfluous metal removed.
 - d. Brackets for framework, fixing devices for panels and anchor bolts to structural background shall be non-corrodible and adequately protected against corrosion. All ferrous members shall be protected by galvanizing.
 - e. Position metalwork accurately, plumb, level and true to line. Fix securely to prevent pulling down, deflection and other movement during use. Do not distort while tightening or fastening.
 - f. Galvanized articles shall be:-
 - 1) Thoroughly and evenly coated with zinc, and
 - 2) Free from strains, bare spots and other defects.
 - g. Unless otherwise specified, zinc-sprayed coating shall be 0.2mm (nominal) thick and the thickness of hot-dip zinc coating shall be minimum 0.085mm.
 - h. Galvanizing shall be hot-dip galvanizing to B.S. 729 : 1971 (1994). Components to be galvanized after fabrication, unless expressly approved otherwise by the Architect.
 - i. Galvanized articles shall be packed with care for transportation.
 - j. No zinc sprayed coated or hot-dip galvanized items shall be welded or drilled and any subsequent cutting is defined as damage which must be subsequently made good after coating or galvanizing.
 - k. Small areas of galvanized coating damaged by any cutting, drilling or welding after galvanizing shall be made good and treated with two coats of metallic zinc-rich priming paint to B.S. 4652 : 1995 Type 2.
 - l. Where galvanized steelwork is to be built into concrete or plaster, the galvanized surface which is to be left exposed on completion of the work shall be thoroughly protected with a wrapping of polythene and plastic tape or some other protection to avoid attack by cement droppings on the galvanizing. The Contractor shall be responsible for replacing any galvanizing steel work which is attacked by cement or plaster.
5. Chromium Plating
 - a. Chromium plating shall be of the quality described in B.S. 1224 : 1970 (1996) for "Service Condition No. 3" with:-
 - 1) Bright finish, or

- 2) Dull finish, or
- 3) Satin finish as specified.
- b. Satin chrome shall be a non-lustrous polished finish.
- c. Chromium plating shall be provided on a base material:-
- d. Of copper or brass, and
- e. Free from visible defects, such as blisters, pits, unplated areas, cloudy patches, cracks or stains.
- 6. Finishes to Aluminium Alloy
 - a. Finishes to aluminium alloy shall be:-
 - 1) Mill, or
 - 2) Polished, or
 - 3) Clear anodized, or
 - 4) Colour anodized as specified.
 - b. A protective lacquer finish shall be applied to these finishes, where specified.
- 7. Anodizing
 - a. Anodizing shall comply with B.S. 1615 : 1987 (1994) or B.S. 3987 : 1991 and of thickness grade and surface texture, as follows:-
 - 1) Clear anodizing exposed to the interior to be thickness grade AA15 (0.015mm),
 - 2) Clear anodizing exposed to the exterior to be thickness grade AA20 (0.020mm),
 - 3) Colour anodizing to be thickness grade AA25 (0.025mm).
 - b. Proprietary approved hardcoat anodic finish shall be approved coating having superior wearing qualities to the above specified B.S. coatings.
 - c. Exposed aluminium surfaces shall be thoroughly cleaned and given a coating of clear non-fading coloured anodized aluminium of the hard coat inorganic type compatible with the alloy used. The coating shall be applied over a caustic etched finish and be sealed effectively. Protect surfaces from handling marks until sealers and protective coatings are applied.
 - d. After cleaning and finishing, a clear, non-yellowing colourless coating shall be applied to all surfaces of aluminium. The coating shall be of sufficient thickness of resist damage from alkaline mortar and plaster for the period of the Contract.
 - e. Protective tapes and strippings shall be supplied on all aluminium surfaces and shall not be removed until the final cleaning down of the building.
- 8. Stainless Steel Work
 - a. Except where specifically shown to be polished, all stainless steel work shall be hairline finished. All exposed screws, fastenings, etc., shall be of appropriate materials to the approval of the Architect.
 - b. Work shall be built by a manufacturer regularly engaged in production of architectural metal work whose workmen shall be skilled at fabricating high grade ornamental metal work. The work shall be fabricated and assembled in as large sections in the shop as consistent with shipping and site requirements. All mitres and joints shall form flush, hairline joints and all joints between sections and all site splices shall produce strength to resist deformation and misalignment. The work shall be fabricated and installed with concealed fastenings in as far as practicable, welding of non-ferrous metals and stainless steel shall be on concealed sides to

minimise pitting, discolouration, weld halo, or other surface imperfections after finishing. Welds shall thoroughly penetrate the material and produce a complete fusion of metal. Wire brushes used for shop or field work shall be made of non-corrosive wire. Dissimilar metals shall be insulated against electrolysis or galvanic action.

- c. Components shall be delivered, stored and handled to prevent damage of whatever nature, factory-finished metals shall be protected in such an approved manner during transportation and installation and such protection shall be removed when directed. Finished surfaces of work shall be cleaned to leave free of stains, markings and defects of any kind and burrs and sharp corners shall be removed from exposed work.

9. Aluminium Work

Aluminium units shall be framed with corners of adequate strength and design to prevent distortion, reinforced with steel as necessary and with hardwood cores where required and factory welded, rivetted or bolted.

10. Contact of Dissimilar Metals

- a. To prevent bi-metallic corrosion, where possible, avoid contact in the completed work between the following metals:-
 - 1) Aluminium alloys and copper alloys, nickel, lead or stainless steel,
 - 2) Iron and steel and copper alloys,
 - 3) Zinc (including galvanizing) and copper alloys or nickel.
- b. Where unavoidable, apply contact surfaces with black bitumen-based coatings to B.S. 3416 : 1991, protective tape or other approved means.

11. Contact of Aluminium and Concrete, etc.

- a. Where possible, avoid contact in the completed work between aluminium and:-
 - 1) Concrete, or
 - 2) Mortar, or
 - 3) Plaster, or
 - 4) Similar materials.
- b. Where unavoidable, apply one coat of black bitumen-based coating on concealed surface or use approved tape.

12. Fixing Generally

- a. Position metalwork accurately, plum, level and true to line or as specified or shown on drawings. Secure components with approved fixings to prevent pulling away from fixing points, excessive deflection or other movement in use. Align fixings and do not distort components locally by overtightening fixings.
- b. Provide suitable lifting gear as required to handle components into position.

13. Concrete Anchors

Where shown, steel or timber work shall be fixed to concrete surfaces by means of Rawbolts, Redheads or similar expanding bolt fasteners or Ramset fasteners. The positioning of all such fixings must be to the approval of the Architect.

14. Adjusting

On completion, lubricate and adjust all moving components and leave in perfect working condition to the satisfaction of the Architect.

15. Galvanized Steel Tubing

Carefully notch, fit and weld galvanized steel tubing to produce accurate joints which shall be ground smooth and treated with two coats of zinc rich priming paint to B.S. 4652 : 1995 Type 2.

16. Castings

Castings shall be sound free from bubbles, cracks or other defects and to include the construction of patterns and moulds as required.

17. Slotted Steel Angle

Cut slotted steel angle square and securely bolt together using nuts, bolts and washers and angle braces as necessary.

18. Fixing Steel Mesh to Framing

Fix steel mesh at 75mm centres as follows:-

- a. To steel framing by tack welding or with 2mm galvanized typing wire.
- b. To wood framing with 1.8 x 25mm galvanized staples.

3.04 TOLERANCES
(NOT USED)

3.05 CLEANING
(NOT USED)

3.06 PROTECTION

- A. Store metalwork components and fittings strictly as recommended by the fabricator of the components and manufacturer of the finishes.
- B. Prevent distortion of metalwork during transit, handling, storage and fixing.
- C. Protect all decorative finishes to metalwork against damp, scratching and other damage.
- D. Apply a strippable coating or masking tape to all stainless steel, anodized aluminium or similar surfaces to prevent contact with mud, ashes, plaster, cement, paint spatter, dust, etc. and only remove as and when necessary for construction or just prior to inspection of handover.
- E. Wash down surplus adhesive with non-damaging solvent or a suitable detergent recommended by the fabricator of the component.
- F. Clean down all metalwork, removing all concrete, plaster, mortar or paint stains and leave the whole perfectly clean on completion. Replace any damaged metalwork.

END OF SECTION